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## PERSONAL STATEMENT

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For more than thirty years, my work has followed two parallel paths in industry and academia, with each informing the other. In industry, I have conducted applied research, designed intelligent systems, and built and led R&D teams across applications including finance, public safety, and automotive systems. Practical challenges encountered in this work have repeatedly shaped my academic research questions, while academic research has provided methods that I have brought back into applied settings.

My academic training includes a B.Sc. from the Technion, an M.Sc. in Computer Science from the Weizmann Institute, and a Ph.D. in Computer Science from Tel Aviv University. My doctoral research, conducted within the research group of Nobel laureate Prof. Leon N. Cooper, combined machine-learning and signal-processing methods for detecting underground structures and tunnels using low-energy acoustic pulses. I am currently a Senior Lecturer at the Holon Institute of Technology, where I conduct research within my Pragmatic AI program.

In industry, my roles at Amdocs, Citigroup, Motorola Solutions, and Questar included applied research, system design, and leadership of AI and software R&D under demanding computational and operational constraints. As an industry partner, I also led research teams preparing and executing competitively funded R&D programs, including a BIRD Israel-US binational grant and an Israel Innovation Authority Magnetron academia-industry grant, with responsibility for the technical work and project milestones.

Energy- and cost-efficient computation has been a recurring theme across both my industry and academic work. At Amdocs, for example, I developed CPU-load models for cost-efficient system sizing, contributing to approximately \$10 million in savings. In academia, this theme has included selective inference in CalexNet and energy-efficient remote-sensing measurements. My recent *Joule Point* project extends this work to GPU inference by identifying energy-optimal operating points. It has already produced a measurement framework, analysis pipeline, and unique dataset, which the proposed grant will extend into predictive models and a decision-support toolkit for energy-efficient AI in data centers.

## EDUCATION

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| 2012–2013 | Post-doctoral Fellow, Computer Vision, Weizmann Institute of Science.                                                                             |
| 2007–2012 | Ph.D., Computer Science (Machine Learning and Signal Processing), Tel-Aviv University, School of Computer Science.                                |
| 2005–2006 | M.Sc., Management (Information System Management), Polytechnic Institute of NYU, Israel Extension, Dept. of Technology Management and Innovation. |

2001–2004 M.Sc., Computer Science (Computer Vision), Weizmann Institute of Science, Faculty of Mathematics and Computer Science.

1991–1995 B.Sc., Computer Science (4-year track), Technion, Israel Institute of Technology, The Taub Faculty of Computer Science.

*Doctoral research supported by the U.S. Army Research Office under "Detection of Underground Installments in Hostile Environment" (PI: Prof. Leon N. Cooper, Nobel Laureate in Physics, Brown University).*

## ACADEMIC EXPERIENCE

|              |                                                        |                  |
|--------------|--------------------------------------------------------|------------------|
| 2024–present | Holon Institute of Technology (HIT), Computer Science. | Senior Lecturer  |
| 2013–present | Afeka College of Engineering, Intelligent Systems.     | M.Sc. Adviser    |
| 2022–present | Bar-Ilan University, Engineering.                      | Adjunct Lecturer |
| 2013, 2015   | Tel-Aviv University, Computer Science.                 | Adjunct Lecturer |
| 2011–2012    | Academic College of Tel Aviv-Yaffo, Computer Science.  | Adjunct Lecturer |

## PROFESSIONAL EXPERIENCE

| Year      | Role                        | Company                                     | Application                                      |
|-----------|-----------------------------|---------------------------------------------|--------------------------------------------------|
| 2020–2024 | Head, Data Science          | Questar Automotive                          | <i>AI for predictive maintenance</i>             |
| 2019–2020 | VP, Artificial Intelligence | Harmon.IE                                   | <i>AI for enterprise document management</i>     |
| 2015–2019 | Head, AI Research; DMTS     | Motorola Solutions Israel Innovation Center | <i>AI for public safety</i>                      |
| 2013–2015 | AVP, Data Science           | Citigroup TLV Innovation Lab                | <i>ML for trading-flow analytics</i>             |
| 2010–2011 | Data Scientist              | Amdocs                                      | <i>Capacity modeling for distributed systems</i> |
| 2006–2009 | Systems Engineer            | Comverse                                    | <i>Personalized content delivery</i>             |
| 2001–2005 | Chief Architect             | McGraw-Hill (Tegrity)                       | <i>Vision-based education multimedia</i>         |

## PUBLICATIONS

### *Peer-reviewed journal and conference papers*

1. Aperstein, Y.; Tzahar, A.; Gottlib, A.; Verber, T.; Shagan Damti, R.; **Apartsin, A.** Selective multipathology chest X-ray classification via rejection mechanisms. Scientific Reports, 2026. [doi.org/10.1038/s41598-026-66294-7](https://doi.org/10.1038/s41598-026-66294-7)
2. Aperstein, Y.; Moran, E.; **Apartsin, A.** IRC-Bench: recognizing entities from contextual cues. Machine Learning and Knowledge Extraction, 8(7), 2026. [doi.org/10.3390/make8070186](https://doi.org/10.3390/make8070186)

3. Atia, T.; Aperstein, Y.; **Apartsin, A.** SeaAlert: robust severity classification and LLM-based information extraction for maritime distress communications. *IEEE Access*, 14, 2026. [doi.org/10.1109/access.2026.3709004](https://doi.org/10.1109/access.2026.3709004)
4. Adamenko, I.; Ben-Aharon, O.; Aperstein, Y.; **Apartsin, A.** Mapping license-plate recoverability under extreme viewing angles. *AI*, 7(7), 2026. [doi.org/10.3390/ai7070237](https://doi.org/10.3390/ai7070237)
5. Aperstein, Y.; **Apartsin, A.** CalexNet: soft cascade-aligned training and calibration for lightweight early-exit branches. *Electronics*, 15, 2026. [doi.org/10.3390/electronics15102149](https://doi.org/10.3390/electronics15102149)
6. Werthaim, M.; Kimhi, M.; **Apartsin, A.**; Aperstein, Y. A benchmark for evaluating diagnostic questioning efficiency of LLMs in patient conversation. *Scientific Reports*, 16, 2026. [doi.org/10.1038/s41598-026-37022-y](https://doi.org/10.1038/s41598-026-37022-y)
7. Aperstein, Y.; Gottlib, A.; Benita, G.; **Apartsin, A.** Explainable semantic text relations. *Information*, 16(12), 2025. [doi.org/10.3390/info16121090](https://doi.org/10.3390/info16121090)
8. Aperstein, Y.; Cohen, Y.; **Apartsin, A.** Generative AI-based platform for deliberate teaching practice. *Education Sciences*, 15(4), 2025. [doi.org/10.3390/educsci15040405](https://doi.org/10.3390/educsci15040405)
9. **Apartsin, A.**; Cooper, L.N.; Intrator, N. Energy-efficient time-of-flight estimation in the presence of outliers. *IEEE JSTARS*, 7(4), 2014. [doi.org/10.1109/jstars.2013.2295324](https://doi.org/10.1109/jstars.2013.2295324)
10. **Apartsin, A.**; Cooper, L.N.; Intrator, N. Time-of-flight estimation in the presence of outliers, Part II. *IEEE Trans. Geoscience and Remote Sensing*, 52(7), 2013. [doi.org/10.1109/tgrs.2013.2276919](https://doi.org/10.1109/tgrs.2013.2276919)
11. **Apartsin, A.**; Cooper, L.N.; Intrator, N. Time-of-flight estimation in the presence of outliers, Part I. *IEEE Trans. Geoscience and Remote Sensing*, 52(6), 2013. [doi.org/10.1109/tgrs.2013.2272737](https://doi.org/10.1109/tgrs.2013.2272737)
12. **Apartsin, A.**; Cooper, L.N.; Intrator, N. Semi-coherent time-of-arrival estimation using regression. *J. Acoustical Society of America*, 132(2), 2012. [doi.org/10.1121/1.4730885](https://doi.org/10.1121/1.4730885)
13. **Apartsin, A.**; Intrator, N. A data-fusion and multiple-ping method for low-power acoustic and seismic sensing. *J. Acoustical Society of America*, 124(4), 2008. [doi.org/10.1121/1.4783265](https://doi.org/10.1121/1.4783265)
14. **Apartsin, A.**; Ferapontova, E.; Gurvich, V. A circular graph: counterexample to the Duchet kernel conjecture. *Discrete Mathematics*, 178, 1998. [doi.org/10.1016/S0012-365X\(97\)81830-4](https://doi.org/10.1016/S0012-365X(97)81830-4)
15. **Apartsin, A.**; Stein, H.; Reiter, G.; Williams, K.; Moscovich, N. Early detection of engine anomalies for AI-based Integrated Vehicle Health Management. *SAE WCX World Congress*, 2022. [doi.org/10.4271/2022-01-0225](https://doi.org/10.4271/2022-01-0225)
16. **Apartsin, A.**; Reiter, G.; Williams, K. AI-based signal-integrity monitoring for integrated vehicle health management. *ELIV*, 2021. [doi.org/10.51202/9783181023846-143](https://doi.org/10.51202/9783181023846-143)
17. Cohen, O.; **Apartsin, A.**; Alon, J.; Katz, E. Robust motion compensation for forensic analysis of egocentric video. *IEEE ICSEE*, 2018. [doi.org/10.1109/icsee.2018.8646211](https://doi.org/10.1109/icsee.2018.8646211)
18. Efrat, N.; Glasner, D.; **Apartsin, A.**; Nadler, B.; Levin, A. Accurate blur models vs. image priors in single-image super-resolution. *IEEE ICCV*, 2013. [doi.org/10.1109/iccv.2013.352](https://doi.org/10.1109/iccv.2013.352)
19. **Apartsin, A.**; Cooper, L.N.; Intrator, N. Biosonar-inspired source localization in low SNR. *BIOSIGNALS / BIOSTEC*, 2011. [doi.org/10.5220/0003126803990404](https://doi.org/10.5220/0003126803990404)
20. **Apartsin, A.**; Cooper, L.N.; Intrator, N. SNR-dependent filtering for time-of-arrival estimation in high noise. *IEEE MLSP*, 2010. [doi.org/10.1109/mlsp.2010.5588848](https://doi.org/10.1109/mlsp.2010.5588848)
21. Galun, M.; **Apartsin, A.**; Basri, R. Multiscale segmentation by combining motion and intensity cues. *IEEE CVPR*, 2005. [doi.org/10.1109/cvpr.2005.244](https://doi.org/10.1109/cvpr.2005.244)

### ***Preprints and submitted***

22. **Apartsin, A.**; Aperstein, Y. The Joule Point: an energy-optimal operating point for AI inference. Preprint, September 2026. DOI: [10.13140/RG.2.2.20385.36964](https://doi.org/10.13140/RG.2.2.20385.36964).
23. **Apartsin, A.**; Aperstein, Y. Three generations of healthcare IT: from the digital record to the computable care process. 2026. [arXiv:2608.08806](https://arxiv.org/abs/2608.08806)
24. **Apartsin, A.**; Aperstein, Y. Clinical communication processing with models trained on LLM-generated synthetic data. 2026. [arXiv:2608.05993](https://arxiv.org/abs/2608.05993)
25. **Apartsin, A.**; Aperstein, Y. Modeling normal is all you need: anomaly detection in multimodal cyber-physical systems. 2026. [arXiv:2607.06094](https://arxiv.org/abs/2607.06094).
26. **Apartsin, A.**; Aperstein, Y. Framing, judging, steering: an assessable competency model for reasoning with generative AI. 2026. [arXiv:2606.05983](https://arxiv.org/abs/2606.05983)
27. **Apartsin, A.**; Aperstein, Y. CoEval: ranking language models for custom tasks without labeled data. 2026. [arXiv:2606.03650](https://arxiv.org/abs/2606.03650)
28. Aperstein, Y.; **Apartsin, A.** Domain-gated latent diffusion: inverse design of HMX-class energetic materials. 2026. [arXiv:2605.26540](https://arxiv.org/abs/2605.26540)
29. **Apartsin, A.**; Sason, O.; Aperstein, Y. Toward a benchmark for controllable simulation of imperfect students with LLMs. 2026. [arXiv:2605.25601](https://arxiv.org/abs/2605.25601)
30. **Apartsin, A.**; Meshulam, Y.; Aperstein, Y. Communication-free collaborative filtering for decentralized multi-robot task allocation. 2026. [arXiv:2605.25584](https://arxiv.org/abs/2605.25584).
31. Aperstein, Y.; **Apartsin, A.** A controlled synthetic benchmark for educational aspect-based sentiment analysis. 2026. [arXiv:2605.25502](https://arxiv.org/abs/2605.25502)
32. Cohen, M.; Shani, S.; Menahem, E.; Aperstein, Y.; **Apartsin, A.** Classifying resume seniority with large language models. [arXiv:2509.09229](https://arxiv.org/abs/2509.09229), 2025.
33. Aperstein, Y.; Gottlib, A.; Benita, G.; **Apartsin, A.** Towards knowledge-aware document systems. [arXiv:2509.08304](https://arxiv.org/abs/2509.08304), 2025.
34. Dahary, S.; Edana, A.; **Apartsin, A.**; Aperstein, Y. Emotion estimation in pop-song lyrics. [arXiv:2509.05617](https://arxiv.org/abs/2509.05617), 2025.
35. Cohen, Y.; Ohayon, D.; Somkin, R.; Aperstein, Y.; **Apartsin, A.** Code review without borders: synthetic vs. real data. [arXiv:2509.04810](https://arxiv.org/abs/2509.04810), 2025.
36. Bolshinsky, I.; Kupiec, S.; Sasson, S.; Aperstein, Y.; **Apartsin, A.** Do large language models need intent? [arXiv:2509.05006](https://arxiv.org/abs/2509.05006), 2025.
37. Cohen, D.; Efrosman, I.; Aperstein, Y.; **Apartsin, A.** Panoramic incident summaries from body-worn footage. [arXiv:2509.04370](https://arxiv.org/abs/2509.04370), 2025.
38. Goren, Y.; Cohen, Y.; **Apartsin, A.**; Aperstein, Y. Interjection classification for human-computer interaction. [arXiv:2509.03181](https://arxiv.org/abs/2509.03181), 2025.

### **PATENTS AND PATENT APPLICATIONS**

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1. Kovalets, S.; Barabanov, S.; Shalev, Y.; **Apartsin, A.** System and method for model-configuration selection. [US 11,868,899 B2](https://patents.google.com/patent/US11868899B2), 2024.
2. Kahlon, H.; **Apartsin, A.**; Lev, D.; Mordel, T. Response to an audio query determined to have a public-safety impact. [US 10,825,450 B2](https://patents.google.com/patent/US10825450B2), 2020.

3. Basri, R.; Brestel, C.; Galun, M.; **Apartsin, A.** Matching portions of input images. [US 8,175,412 B2](#), 2012.
4. Stein, Y.; Vardi, Y.; **Apartsin, A.** Unification and optimization of machine-learning inference pipelines. [US 2024/0013905 A1](#), 2024.
5. **Apartsin, A.**; Henrichs, K.; Kossaczky, I.; Kovalets, S.; Mavliutov, Y.; Korchahin, I. DTC rulebook generation system and method. [US 2024/0338980 A1](#), 2024.
6. Shalev, Y.; Henrichs, K.; Kossaczky, I.; **Apartsin, A.** Determining a score for a driver within a fleet. [IL 308506A](#), 2025.
7. Henrichs, K.; Blanc, S.M.; Shalev, Y.; **Apartsin, A.** Model training for a new signal from metadata representations of known signals. [US 2023/0351160 A1](#), 2023.
8. Bakalo, R.; **Apartsin, A.**; Stein, Y.; Vardi, Y. Feature-selection recommendation. [US 2022/0383141 A1](#), 2022.
9. **Apartsin, A.**; Stein, Y.; Vardi, Y. Physical model-based machine learning. [US 2022/0382939 A1](#), 2022.
10. **Apartsin, A.**; Tchemerisov, V.; Cooperman, V. Search user interface. [US 2009/0070321 A1](#), 2009.